9.4.1 It should be possible to display the comp lete system configuration, the available configuration and the configuration in use.

9.4.2 The INS should provide the means to display the type of data, source and availability.

9.4.3 The INS should provide the means to displa y the type of function and availability.

RESOLUTION MSC.252(83)
(adopted on 8 October 2007)
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9.4.4 The INS should provide the means to display the device identification and its availability.

9.4.5 Ships and system related parameters and settings should be displayed on demand.

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9.4.4 The INS should provide the means to display the device identification and its availability.

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10 Human Machine Interface

10.1 General

10.1.1 For the design and layout of human mach ine interface (HMI) of the INS, MSC/Circ.982 and relevant guidance on application of SOLAS regulation V/15 adopted by the Organization should be taken into account.

10.1.2 Integrated graphical and alphanumeric display and control functions should adopt a consistent human machine interface (HMI) philosophy and implementation.

10.1.3 The design and implementation of the INS s hould ensure that it is simple to operate by a trained user.

10.2 System Design

10.2.1 The design of the system should facilitate the tasks to be performed by the bridge team and pilot in navigating the ship safely under all operational conditions.

10.2.2 The configuration of the equipment and presentation of information at workstations should permit observation or monitoring by th e bridge team and pilot under all operating conditions.

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SOURCE: MSC.252(83).pdf
10.2.2 The configuration of the equipment and presentation of information at workstations should permit observation or monitoring by th e bridge team and pilot under all operating conditions.

10.2.3 The design of the system should avoid the potential single point failure by one person during operation, and should minimize the risk of human error.

10.2.4 The operation of the system should be designe d to avoid distraction from the task of safe navigation.

10.3 Display

10.3.1 Information should be presented consistently within and between different sub-systems. Standardized information presentation, symbols and coding should be used according to resolution MSC.191(79).

10.4 Input

10.4.1 The INS should be so designed that th e requested manual inputs are consistent throughout the system and can be easily executed.

10.4.2 The INS should be designed that the ba sic functions can be easily operated.

10.4.3 Complex or error-prone interacti on with the system should be avoided.
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10.4.4 For manual inputs that may cause unint ended results, the INS should request confirmation before acceptance, thus providing a plausibility check.

10.4.5 Checks in the dialogue and in the input ha ndling should be provided to prevent erroneous data or control inputs.

10.4.6 Wherever possible, an “ undo” function should be provided.

11 INS Back-up requirements and redundancies

11.1 General

11.1.1 Adequate back-up arrangements should be provi ded to ensure safe navigation in case of a failure within the INS.

11.1.1.1 In case of failure of one part or functi on of the INS, including network failures, it should be possible to operate each other individua l part or function separately; at least the requirements specified for individual equipment adopted by the Organization should be met, as far as applicable.

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SOURCE: MSC.252(83).pdf
should be possible to operate each other individua l part or function separately; at least the requirements specified for individual equipment adopted by the Organization should be met, as far as applicable.

11.1.1.2 The back-up arrangement should enable a safe take-over of the INS functions and ensure that an INS failure does not result in a critical situation.

11.1.2 The failure of a single task station should not result in the loss of a function mandated by the carriage requirements of SOLAS.

11.1.3 In case of a breakdown of one task station, at least one task station should be able to take over the tasks.

11.1.4 The failure or loss of one hardware component of the INS should not result in the loss of any one of the INS tasks:

Route planning
Route monitoring
Collision avoidance
Navigation control data
Status and data display
Alert management.

Where track control is an INS function, this woul d not require the duplication of heading control or autopilot.

11.1.5 The INS should allow that the back-up compone nt automatically (if possible) takes over the operation of the primary component.
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or autopilot.

11.1.5 The INS should allow that the back-up compone nt automatically (if possible) takes over the operation of the primary component.
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11.2 Hardware Redundancies

11.2.1 Required navigational sensor/source back-up

For the following sensors/sources of an INS, an approved back-up should be available for the INS:

electronic position fixing
heading measurement
speed measurement
radar
chart database.

12 System failures and fallback arrangement

12.1 The INS should, after a failure, and when the back-up activation is not successful support the availability of esse ntial information and f unctions through the use of appropriate fallback arrangements (see 12.7).

12.2 Restored operation